Rational Periodic Multiplier Moduli under Good Reduction: A Hénon Certificate and Exact Audit

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Abstract

Periodic multipliers of a symplectic map occur in reciprocal pairs, but reciprocity alone does not prevent an eigenvalue from having a prescribed rational modulus. We isolate an arithmetic obstruction for monic area-preserving generalized Hénon maps. For a finite composition defined over an S-integer ring, every finite complex periodic point is S-integral, every return multiplier is an algebraic S-unit, and an exact rational modulus q can involve only the rational primes declared bad by S. The modulus statement does not assume that the multiplier is rational. Its proof passes to a Galois closure and all places above the rational bad set, so complex conjugation preserves the relevant unit group; it never identifies complex conjugation with reciprocal pairing.

For the frozen map

$$H_u(X,Y) = (X^2 - u - Y, X), \qquad u^3 - 2u^2 + 2u - 2 = 0,$$

with u the unique real root, the bad set is empty. Hence every exactly rational periodic multiplier modulus equals one at every period, excluding an exact rational-prime modulus clock. A source-locked exact-arithmetic audit through periods $1, 2, 3$ removes lower-period branches, recovers five cycles, verifies determinant-one monodromy and reciprocal unit polynomials, and finds one unit-modulus cycle and four cycles with irrational algebraic-unit moduli. This finite ledger audits the software; it is not the source of the all-period result. The control $a = -15/16$ has fixed point $(5/4, 5/4)$ and multipliers $2, 1/2$, showing that the bad-prime support bound is sharp. The conclusion is a narrow good-reduction certificate, not a universal obstruction for symplectic dynamics or for irrational instability rates.

1 Introduction

Polynomial automorphisms of the affine plane provide a particularly explicit meeting point of complex, arithmetic, and conservative dynamics. Generalized Hénon maps form the basic nonlinear class in the structural description of plane polynomial automorphisms (Friedland and Milnor, 1989). Their arithmetic has been studied through periodic points, local and canonical heights, and reduction at non-Archimedean places (Silverman, 1994; Marcello, 2003; Kawaguchi, 2006, 2013; Ingram, 2014; Allen et al., 2018). Recent work also emphasizes rational periodic points in families and arithmetic or multiplier rigidity (Hsia and Kawaguchi, 2018; Kim et al., 2024; Cantat and Dujardin, 2026b,a).

This paper asks a deliberately narrower question. Let λ be an eigenvalue of the derivative of a return map along a finite complex periodic orbit. When can its exact complex modulus be a positive

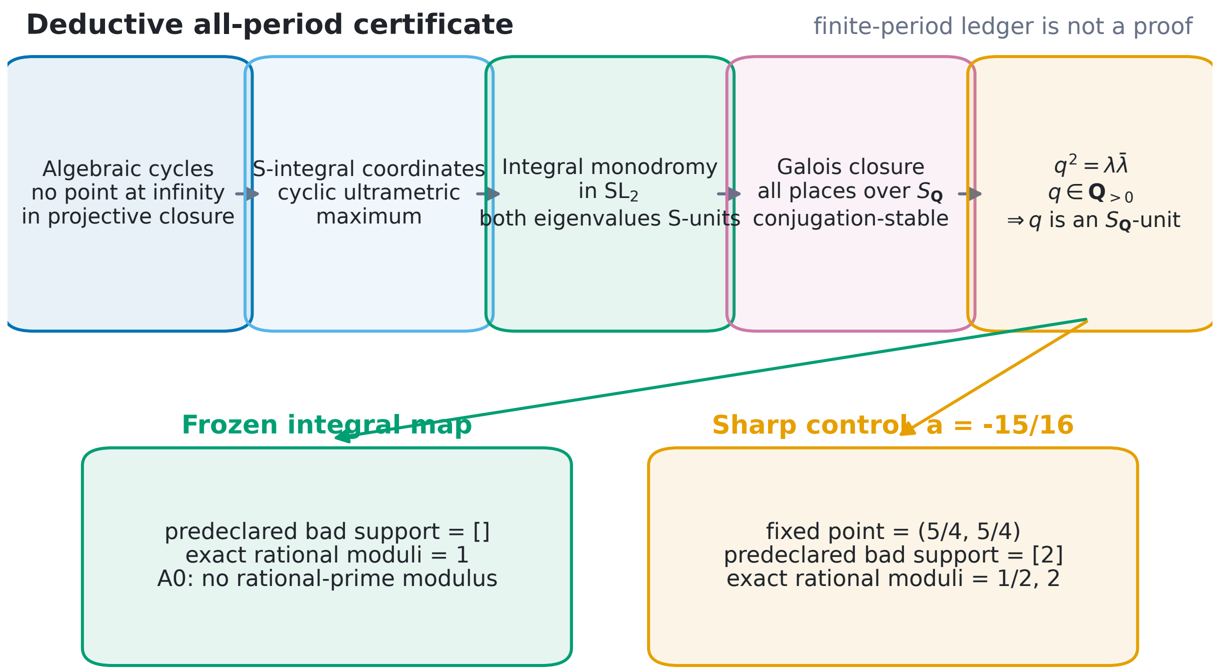

Figure 1: The all-period result is deductive. Algebraicity, a cyclic non-Archimedean maximum, integral special-linear monodromy, and a conjugation-stable Galois closure yield rational bad-prime support. The frozen integral map has empty finite bad set and therefore only rational modulus 1; the planted denominator-2 control realizes 1/2 and 2. All reported candidate and control values are read from the frozen audit package.

rational number—and in particular a rational prime? The qualifier *exact* matters: a decimal close to an integer is not an arithmetic equality. So does the distinction between $\lambda \in \mathbb{Q}$ and $|\lambda| \in \mathbb{Q}$. An algebraic multiplier may be nonreal and nonrational while its modulus is rational.

For determinant-one maps the two eigenvalues are reciprocal, yet this geometric fact alone gives no rationality obstruction: the pair $(2, 1/2)$ is already reciprocal. The additional input here is good reduction. A cyclic non-Archimedean maximum argument makes periodic coordinates integral; integral special-linear monodromy then makes both multipliers units. To pass from a unit statement about λ to a support statement about $|\lambda|$, one must also handle complex conjugation correctly. Figure 1 summarizes this dependency chain.

Our contribution has three parts.

1. We prove an all-period theorem for finite compositions of monic, area-preserving generalized Hénon maps over $\mathcal{O}_{K,S}$: finite periodic coordinates are S-integral, return multipliers are algebraic S-units, and every exact rational modulus is supported on the rational primes below S.

2. We specialize the theorem to one source-locked algebraic-integer parameter u. The resulting global polynomial symplectic automorphism has no periodic multiplier of exact rational-prime modulus at any period, although irrational algebraic-unit moduli larger than one remain possible.

3. We supply a reproducible exact-arithmetic audit through period three, together with a sharp positive control and boundary controls. The audit tests the implementation of exact branches, monodromy, elimination, and modulus classification; it does not extrapolate a finite search to all periods.

The result should not be read as a universal symplectic no-go theorem. Symplectic maps with rational multiplier moduli other than one exist, including the elementary control in Section 5. Nor do we constrain irrational algebraic moduli, singular values, or Lyapunov exponents. The obstruction concerns one exact arithmetic target under explicitly declared good-reduction hypotheses.

2 Arithmetic and multiplier context

The literature surrounding Hénon dynamics is much broader than the support certificate proved here. The classical plane-automorphism framework supplies the geometric setting (Friedland and Milnor, 1989); arithmetic studies use heights and local filtrations to control or organize orbits (Silverman, 1994; Marcello, 2003; Kawaguchi, 2006, 2013; Ingram, 2014). Non-Archimedean Hénon systems also exhibit their own attractors and horseshoes (Allen et al., 2018). Family-level work studies periodic parameters and rational cycles (Hsia and Kawaguchi, 2018; Kim et al., 2024). Separately, dynamical Manin–Mumford and recent multiplier-rigidity results impose powerful global constraints under hypotheses very different from ours (Dujardin and Favre, 2017; Cantat and Dujardin, 2026b,a).

Table 1 locates the present statement. We use a standard good-reduction mechanism but package it around an exact rational *modulus*, rather than rationality of the multiplier itself. The main technical caution is that the chosen complex conjugate $\bar{\lambda}$ need not equal the reciprocal return eigenvalue λ^{-1}. Both operations occur in the proof, but at different stages.

Table 1: Position of the present result relative to adjacent Hénon theory.

Theme	Typical object or tool	Boundary relative to this paper
Plane automorphisms	Structural and complex dynamics of generalized Hénon maps	Supplies the geometric class; does not by itself force rational-modulus support.
Arithmetic dynamics	Heights, local filtrations, periodic points, and reduction	Supplies the natural good-reduction setting; our theorem isolates a short multiplier-support consequence.
Rational periodic points	Existence and abundance in maps or families	Concerns coordinates of cycles, not the exact modulus target studied here.
Multiplier rigidity	Spectra, fields of definition, or recovery of maps	Global rigidity questions use different hypotheses; we prove a local arithmetic support certificate.
This paper	Cyclic integrality plus SL_2 monodromy	If $\lvert\lambda\rvert \in \mathbb{Q}_{>0}$, only predeclared rational bad primes can occur.

3 A good-reduction support theorem

Let K be a number field, let S be a finite set of places containing the archimedean places, and put $R = \mathcal{O}_{K,S}$. Write $\overline{R}$ for the integral closure of R in $\overline{K}$. Let $S_{\mathbb{Q}}$ be the finite set of rational primes below the finite places in S. For $1 \leq i \leq m$, take a monic polynomial $p_i \in R[X]$ of degree $d_i \geq 2$

and define

$$H_i(X,Y) = (p_i(X) - Y, X), \qquad F = H_m \circ \cdots \circ H_1. \tag{1}$$

Each factor has polynomial inverse $H_i^{-1}(X,Y) = (Y, p_i(Y) - X)$ and derivative

$$DH_i(X,Y) = \begin{pmatrix} p_i'(X) & -1 \\ 1 & 0 \end{pmatrix}, \qquad \det DH_i = 1. \tag{2}$$

Thus F is a global polynomial automorphism preserving $dX \wedge dY$.

Theorem 3.1 (Rational-modulus support)**.** *Every finite complex periodic point of F has coordinates in $\overline{R}$. If $F^n(P) = P$, both eigenvalues of $D_P F^n$ are units of $\overline{R}$. Fix the given embedding into $\mathbb{C}$. If one eigenvalue λ satisfies $|\lambda| = q \in \mathbb{Q}_{>0}$, then*

$$q \in \mathbb{Z}[S_{\mathbb{Q}}^{-1}]^{\times}_{>0} = \left\{ \prod_{p \in S_{\mathbb{Q}}} p^{e_p} : e_p \in \mathbb{Z} \right\}. \tag{3}$$

In particular, an exact rational-prime modulus forces that prime to belong to $S_{\mathbb{Q}}$; no rationality assumption on λ is required.

The proof is split into four lemmas to make the logical dependencies explicit.

Lemma 3.2 (Algebraicity)**.** *Every finite complex periodic point of F is algebraic over K.*

Proof. Let $F^n(P) = P$ and expand the orbit through its m individual factors during each of the n returns. The successive first coordinates form a cyclic sequence $(z_j)_{j \in \mathbb{Z}/N\mathbb{Z}}$, with $N = mn$, satisfying

$$p_{i_j}(z_j) - z_{j+1} - z_{j-1} = 0, \tag{4}$$

where i_j is periodic modulo m. Homogenize these N equations with one coordinate Z. On the hyperplane $Z = 0$, monicity and $d_{i_j} \geq 2$ reduce the jth equation to $Z_j^{d_{i_j}} = 0$. Hence all Z_j vanish, which is impossible in projective space. The projective zero set therefore misses the hyperplane at infinity.

A positive-dimensional projective subvariety meets every hyperplane, since the hyperplane class is ample. Thus the projective zero set is zero-dimensional, and the affine cyclic system has finitely many complex solutions. Because its equations have coefficients in K, every coordinate of every solution is algebraic over K. □

Lemma 3.3 (Cyclic non-Archimedean maximum)**.** *The coordinates of every finite periodic point belong to $\overline{R}$.*

Proof. Choose a finite extension L/K containing the orbit coordinates. Fix a finite place w of L above a place outside S. The coefficients of each p_i have w-absolute value at most one. Suppose $M = \max_j |z_j|_w > 1$ and choose j attaining the maximum. Since p_{i_j} is monic of degree at least two,

$$|p_{i_j}(z_j)|_w = |z_j|_w^{d_{i_j}} = M^{d_{i_j}} > M. \tag{5}$$

But the recurrence (4) gives

$$|p_{i_j}(z_j)|_w = |z_{j+1} + z_{j-1}|_w \leq \max\{|z_{j+1}|_w, |z_{j-1}|_w\} \leq M,$$

a contradiction. Therefore all orbit coordinates have absolute value at most one at every finite place outside S. Lemma 3.2 and the valuation criterion for integrality imply that they lie in $\overline{R}$. □

Lemma 3.4 (Unit monodromy)**.** *For every P with $F^n(P) = P$, both eigenvalues of $D_P F^n$ are units of $\overline{R}$.*

Proof. By Lemma 3.3, every entry $p_i'(z_j)$ in (2) belongs to $\overline{R}$. The return monodromy is a product of these determinant-one matrices, so

$$M_P = D_P F^n \in \mathrm{SL}_2(\overline{R}), \qquad \chi_P(T) = T^2 - \mathrm{tr}(M_P)T + 1.$$

Both eigenvalues are integral over R. The other eigenvalue is λ^{-1} and is integral by the same characteristic polynomial. Consequently λ and λ^{-1} belong to $\overline{R}$, which is exactly the required unit statement. □

Lemma 3.5 (Conjugation-stable rational support)**.** *If the multiplier in Lemma 3.4 satisfies $|\lambda| = q \in \mathbb{Q}_{>0}$, then (3) holds.*

Proof. Choose a finite Galois extension $M/\mathbb{Q}$ inside $\mathbb{C}$ that contains K, the orbit, and λ. Let T be *all* finite places of M above the rational primes in $S_{\mathbb{Q}}$. Lemma 3.4 implies $\lambda \in \mathcal{O}_{M,T}^{\times}$. The set T is stable under $\mathrm{Gal}(M/\mathbb{Q})$ because it is defined by underlying rational primes. Complex conjugation therefore sends T-units to T-units, so $\bar{\lambda} \in \mathcal{O}_{M,T}^{\times}$ and

$$q^2 = |\lambda|^2 = \lambda\bar{\lambda} \tag{6}$$

is a rational T-unit. For every rational prime $\ell \notin S_{\mathbb{Q}}$, valuation of (6) gives $2v_\ell(q) = 0$. Thus $v_\ell(q) = 0$, proving (3). □

The four lemmas prove Theorem 3.1. The passage to all places above $S_{\mathbb{Q}}$ is essential: the original field K and an arbitrary chosen set of its places need not be stable under complex conjugation. Just as importantly, equation (6) does *not* identify $\bar{\lambda}$ with the reciprocal eigenvalue. Reciprocity proves that λ^{-1} is an S-unit in Lemma 3.4; conjugation proves that $\bar{\lambda}$ is a T-unit in Lemma 3.5. These are separate facts.

Corollary 3.6 (Integral case)**.** *If S has no finite places, every exact rational periodic multiplier modulus equals 1. If the multiplier itself is rational, it equals $+1$ or -1.*

Proof. The positive units of $\mathbb{Z}$ are just 1. The second statement follows from $|\lambda| = 1$. □

4 The frozen Hénon certificate

Let u be the unique real root of

$$P(U) = U^3 - 2U^2 + 2U - 2. \tag{7}$$

The derivative $P'(U) = 3U^2 - 4U + 2$ is everywhere positive because its discriminant is negative, so the real root is unique. Since P is monic, u is an algebraic integer. We freeze

$$H_u(X,Y) = (X^2 - u - Y, X), \qquad H_u^{-1}(X,Y) = (Y, Y^2 - u - X). \tag{8}$$

The derivative has determinant one, so this is a global polynomial symplectic automorphism of $\mathbb{A}^2$.

Corollary 4.1 (Frozen all-period obstruction)**.** *For every multiplier λ of every finite periodic orbit of H_u,*

$$|\lambda| \in \mathbb{Q}_{>0} \quad \Longrightarrow \quad |\lambda| = 1.$$

Consequently no positive rational prime is a periodic multiplier modulus, even when λ is nonrational.

Table 2: Source-locked exact-period ledger over $\mathbb{Q}[u]/(P)$. Counts are for exact points and cycles after lower-period removal.

n	points	cycles	trace eliminant	multiplier eliminant
1	2	2	$T^2 - 4T - 4u$	$L^4 - 4L^3 + (2-4u)L^2 - 4L + 1$
2	2	1	$T + 4u - 14$	$L^2 + (4u-14)L + 1$
3	6	2	$T^2+(16u-20)T+96u^2-164u+8$	$L^4+(16u-20)L^3+(96u^2-164u+10)L^2+(16u-20)L+1$

Proof. Apply Corollary 3.6 over $K = \mathbb{Q}(u)$ to the monic polynomial $X^2 - u$. $\square$

The conclusion is an obstruction to one exact target, not to instability. An algebraic unit can have an irrational absolute value greater than one; for example, the integer matrix $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has determinant one and spectral radius $(3+\sqrt{5})/2$. Corollary 4.1 also does not decide whether $+1$ or -1 occurs as a multiplier of H_u.

5 Exact audit and sharp controls

The theorem above supplies the all-period result. The computation described here has a different role: it checks that the exact periodic-branch, monodromy, elimination, and modulus-classification software behaves as specified on the first nontrivial periods. The parameter, cutoff, controls, and pass/fail rules were source-locked before candidate execution. No external prime table or zero-ordinate data was accessed.

5.1 Exact-period ledger

For exact periods $n = 1, 2, 3$, the audit builds the cyclic recurrence over $\mathbb{Q}[u]/(P)$, explicitly removes all lower-period branches, computes exact Gröbner bases and resultants, and independently evaluates the monodromy trace. Table 2 records the compact eliminants. The symbol T denotes trace and L a multiplier. Every multiplier polynomial is monic and reciprocal with constant term one.

After taking the norm to $\mathbb{Q}$, the rational-root theorem leaves only $L = \pm 1$ as possible rational roots; exact substitution rejects both at each audited period. To classify moduli without a floating-point rationality test, the implementation uses, for each real-hyperbolic selected cycle, the exact relation

$$M^2 - (T^2 - 2)M + 1 = 0, \qquad M = \lambda\bar{\lambda},$$

and declares M rational only when its irreducible minimal polynomial over $\mathbb{Q}$ has degree one. A separate exact square test is then required before $q = \sqrt{M}$ is called rational. The real-elliptic case is certified directly from conjugate reciprocal roots, which give $M = 1$. Display approximations never enter either decision.

The selected real embedding contains five audited cycles: two fixed cycles, one 2-cycle, and two 3-cycles. One elliptic fixed cycle has exact modulus 1; the remaining four are hyperbolic and have irrational algebraic-unit moduli. Figure 2 reports these counts and the exact software checks. The observed finite rational-modulus set is $\{1\}$, in agreement with Corollary 4.1; agreement is a consistency check, not an inductive proof.

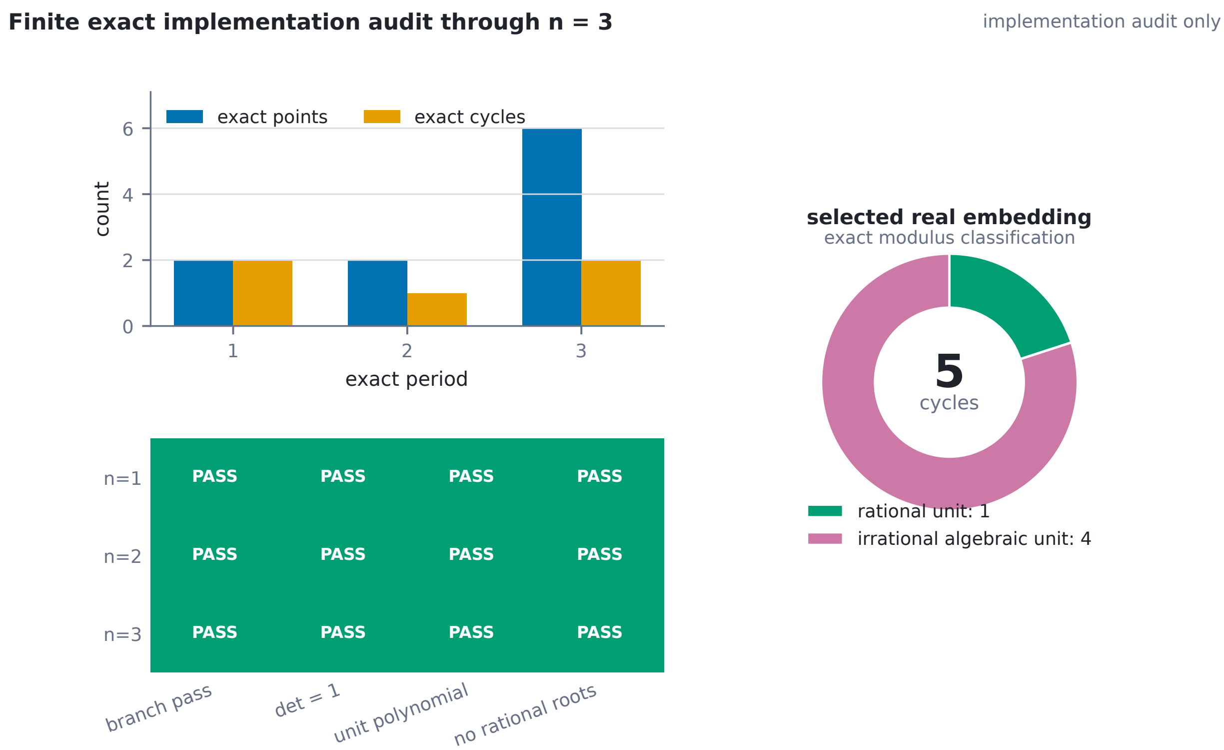

Figure 2: The finite exact implementation audit. Lower-period branch removal gives $2, 1, 2$ cycles at exact periods $1, 2, 3$. Every audited branch passes the determinant, unit-polynomial, and rational-root checks. At the selected real embedding, one of the five cycles has rational unit modulus and four have irrational algebraic-unit moduli.

5.2 Sharp positive and boundary controls

The support conclusion is sharp at the level of allowed primes. Set $a = -15/16$ and $r = 5/4$. The point (r, r) is fixed by $H_a(X, Y) = (X^2 - a - Y, X)$ because $a = r^2 - 2r$. At that point,

$$DH_a(r, r) = \begin{pmatrix} 5/2 & -1 \\ 1 & 0 \end{pmatrix}, \qquad L^2 - \frac{5}{2}L + 1 = (L - 2)(L - 1/2). \tag{9}$$

The coefficient denominator declares bad support $\{2\}$ before multiplier access, and the generic exact-modulus pipeline recovers 2 and 1/2.

Two additional controls delimit the theorem. For $J_{a,\delta}(X, Y) = (X^2 - a - \delta Y, X)$, the derivative determinant is δ. At $(a, \delta) = (0, 2)$, the fixed point $(0, 0)$ has characteristic polynomial $L^2 + 2$ and multipliers $\pm i\sqrt{2}$, which are not algebraic units; the unit conclusion is correctly refused unless the determinant's bad prime is tracked. Conversely, the cat-map example above is integral and determinant one but has an irrational algebraic-unit spectral radius larger than one. It confirms that the rational-support theorem does not bound all instability rates.

The official run executed eight controls and preflight checks before opening the candidate gate, then seven candidate and scope checks. All 15 registered runs passed, as did 39 unit and protocol tests. The computation used exact symbolic arithmetic on a local CPU and no GPU.

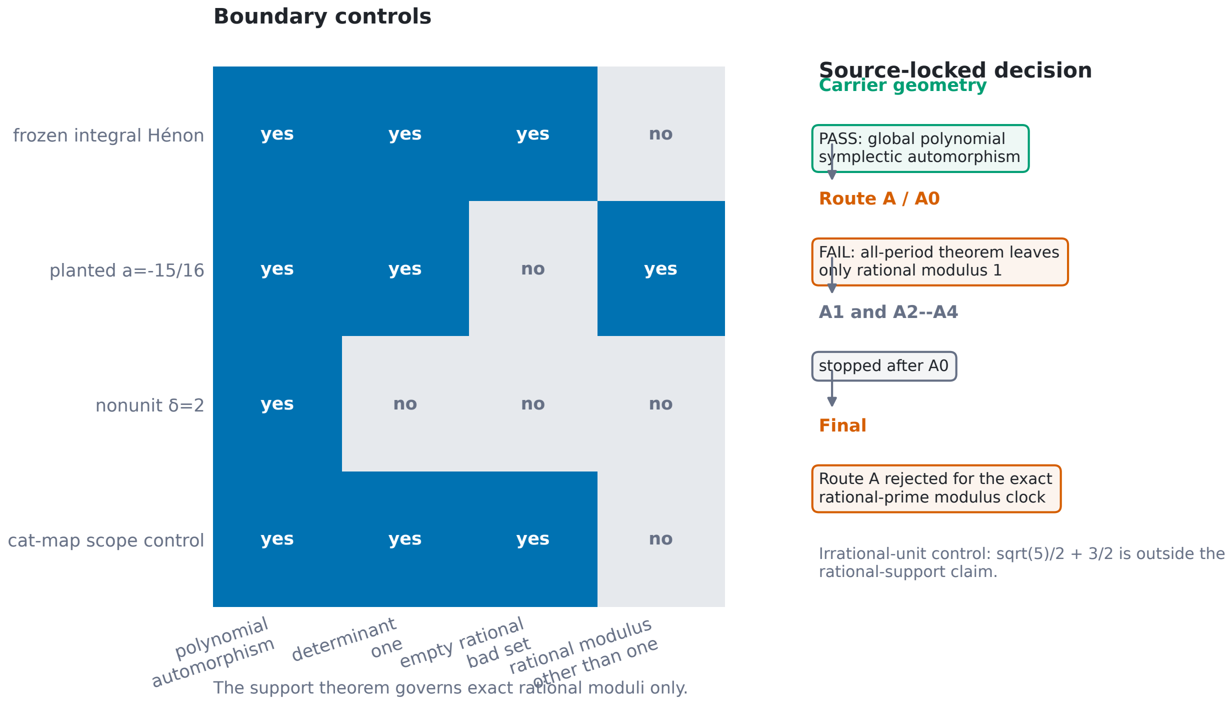

Figure 3: Boundary controls and the source-locked route decision. The frozen map passes the geometric carrier gate but fails the exact rational-prime modulus gate by theorem. The denominator-2 control shows the good-reduction hypothesis is substantive; the nonunit-Jacobian and irrational-unit controls show which nearby conclusions are outside scope.

6 Route decision and scope

The source-locked decision separated carrier geometry from the proposed arithmetic target. Geometry passed: equation (8) is a global polynomial symplectic automorphism. The first arithmetic gate, A0, failed: Corollary 4.1 excludes every exact rational-prime multiplier modulus at every period. Later Route-A gates were therefore stopped by scope, and no alternative route was opened. Figure 3 places this decision beside the controls.

The negative result has precise boundaries.

- It applies to finite complex periodic points of the monic generalized Hénon compositions in Theorem 3.1. Monicity and degree at least two are used both at infinity and in the strict maximum argument.
- It controls exact rational complex moduli. Near-rational decimal values, irrational algebraic moduli, singular values, and Lyapunov exponents are not classified.
- It is not a universal symplectic obstruction. Bad reduction can support rational moduli such as 2 and 1/2, and other symplectic maps need not satisfy the theorem's arithmetic hypotheses.
- It does not assert a prime–orbit correspondence, a spectral determinant identity, compactness, quantization, or the absence of all hyperbolicity. No target prime/zero dataset was consulted.

There is also a controlled extension boundary. If a generalized Hénon factor has determinant an S-unit rather than one, a related unit statement may be recovered by tracking the determinant in the bad set. If its leading coefficient is a nonunit, its valuations must likewise be declared bad and the projective argument must be redone. Neither extension is claimed here.

7 Discussion

The certificate illustrates why a modulus question can be stricter than a multiplier question. Integral monodromy gives a reciprocal algebraic-unit pair, but an exact modulus uses the chosen Archimedean embedding. Passing to a conjugation-stable Galois closure connects that Archimedean equality to non-Archimedean support without assuming the chosen field or place set is itself stable under conjugation. This repair is small algebraically but decisive logically.

The result is intentionally elementary. It does not compete with the deeper height, equidistribution, or rigidity theories cited in Section 2. Its value is as a fail-fast arithmetic certificate: before searching periodic orbits for a prescribed rational modulus, inspect the good-reduction data. When the rational bad set is empty, the search for a rational-prime modulus is closed at every period. When the bad set is nonempty, the theorem supplies a finite support filter rather than an existence theorem.

The sharp control also suggests the right interpretation of "sharp." The theorem predicts which rational primes are allowed to appear, not that every allowed prime is realized for every map or period. Realization is a separate dynamical problem. Likewise, the low-period audit is useful because it tests branch saturation, exact norm calculations, reciprocal polynomials, and modulus elimination; none of those finite checks strengthens the quantified all-period theorem.

8 Conclusion

For monic area-preserving generalized Hénon compositions over an S-integer ring, finite periodic coordinates are S-integral and return multipliers are algebraic S-units. A conjugation-stable Galois-closure argument then forces every exact rational modulus to be supported on the predeclared rational bad primes. At the frozen integral parameter $u^3 - 2u^2 + 2u - 2 = 0$, the support is empty and the only possible rational modulus is one, so the exact rational-prime modulus route is ruled out at all periods. The exact audit through period three and the $a = -15/16$ control verify the implementation and the sharpness boundary. The outcome is a reusable good-reduction screen, with an explicit boundary between deductive arithmetic obstruction and finite computational evidence.

A Reproducibility and evidence separation

The experiment package contains a machine-readable source lock, an exact period ledger, exact polynomial records, candidate and control audits, a scope audit, a negative-result ledger, the complete run registry, and a final hash manifest. The official sequence was:

1. validate the source lock, forbidden-data scanner, proof structure, exact parameter, symplectic identity, and all controls;
2. open the candidate gate only after every prerequisite passed;
3. run the exact-period candidate audit through the locked cutoff $n = 3$;
4. run scope guards and the full 39-test protocol suite;
5. close the artifact set with SHA-256 hashes.

All result-bearing figures are regenerated from those frozen JSON artifacts. The figure loader checks the official run status, candidate identity, failure count, and source-lock hash before writing output. PDF and SVG files are vector masters; PNG files are review renderings only.

The evidence hierarchy is fixed. Theorem 3.1 and Corollary 4.1 come from proof. The $n \le 3$ ledger checks an implementation against exact identities. The five selected-embedding cycles are finite observations. None of the experimental artifacts is cited as a proof of an all-period claim.

B Exact audit identities

At a cyclic orbit $(x_0, \dots, x_{n-1})$, the recurrence is

$$x_{j+1} + x_{j-1} = x_j^2 - u \quad (j \bmod n),$$

and the monodromy is the ordered product

$$M_n = \prod_{j=n-1}^{0} \begin{pmatrix} 2x_j & -1 \\ 1 & 0 \end{pmatrix}.$$

The determinant is identically one before any quotient-ring reduction. Consequently every orbitwise characteristic polynomial has form $L^2 - (\operatorname{tr} M_n)L + 1$. The period-1 and period-3 eliminants in Table 2 combine two cycles and hence have degree four; the period-2 eliminant represents its unique cycle and has degree two.

After taking field norms from $\mathbb{Q}(u)$, the multiplier polynomials over $\mathbb{Q}$ have degrees $12, 6, 12$ for $n = 1, 2, 3$, respectively. Each is monic, reciprocal, and has constant term one. Their exact evaluations at $L = \pm 1$ are nonzero, which certifies the absence of rational multiplier roots through the cutoff. This rational-root audit is strictly weaker than the rational-*modulus* theorem: a nonrational eigenvalue can in principle have rational modulus, which is why the separate $M = \lambda\bar{\lambda}$ elimination is required.

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